

ASSESSMENT TOOL

GOVERNMENT SCHEME ELIGIBILITY AGENT USING RAG

Retrieval-Augmented Generation

Student Name	M. Someswar Reddy
Register Number	192472081

1. Aim

To develop a Retrieval-Augmented Generation (RAG) based Government Scheme Eligibility Agent that retrieves relevant information from government scheme documents and presents eligibility information with supporting retrieved document context.

2. Objective

To build a complete RAG pipeline consisting of document loading, text chunking, embedding generation, FAISS vector-store indexing, similarity retrieval, and grounded eligibility responses for citizen profiles.

3. Problem Statement

Citizens may need to identify government schemes for which they may be eligible. The system retrieves relevant scheme information from the available government documents and displays the supporting retrieved text instead of relying only on general model knowledge.

4. Documents Used

No.	Document	Type
1	Revised PM-KISAN Operational Guidelines (English)	Government PDF
2	PMAY-U 2.0 Scheme Guidelines	Government PDF

Note: The present implementation uses the two uploaded government PDFs. The assessment instruction states a minimum of 3+ government scheme documents; therefore, a third official scheme document should be added before final submission if this requirement is enforced by the faculty.

5. RAG Architecture

Government Scheme PDFs → Document Loading → Text Chunking → Sentence Embeddings → FAISS Vector Store → Similarity Retrieval → Eligibility Agent → Grounded Answer + Retrieved Chunks + Source/Page

6. Design Choices

Component	Selected Design
Embedding model	sentence-transformers/all-MiniLM-L6-v2
Vector store	FAISS
Chunk size	900 characters
Chunk overlap	150 characters
Top-k	5
Retrieval method	Cosine-style similarity using normalized embeddings and FAISS inner product
Document format	PDF

7. Why These Design Choices

- all-MiniLM-L6-v2 provides compact sentence embeddings suitable for semantic similarity retrieval.

- FAISS provides fast vector similarity search over the generated embeddings.
- A 900-character chunk size keeps related eligibility information together while keeping retrieval focused.
- A 150-character overlap helps preserve context between adjacent chunks.
- Top-k = 5 provides several relevant chunks for the answer generation stage.

8. Procedure

1. Place the government PDF documents in the documents folder.
2. Load each PDF page using PyPDF.
3. Extract and clean the text.
4. Split the text into overlapping chunks.
5. Generate embeddings for all chunks.
6. Create a FAISS vector index.
7. Convert the citizen profile or query into an embedding.
8. Retrieve the five most similar chunks.
9. Identify relevant scheme information from the retrieved context.
10. Display the scheme, supporting eligibility/context clause, source and page.
11. Demonstrate answerable, multi-information and out-of-scope queries.

9. Python Program

```
import re
from pathlib import Path

import faiss
import numpy as np
from pypdf import PdfReader
from sentence_transformers import SentenceTransformer

BASE_DIR = Path(__file__).resolve().parent
DOC_DIR = BASE_DIR / "documents"

EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"
TOP_K = 5
CHUNK_SIZE = 900
CHUNK_OVERLAP = 150

def load_documents():
    documents = []

    for pdf_path in DOC_DIR.glob("*.pdf"):
        reader = PdfReader(str(pdf_path))

        for page_number, page in enumerate(reader.pages, start=1):
            text = page.extract_text() or ""
            text = re.sub(r"\s+", " ", text).strip()

            if text:
                documents.append({
                    "source": pdf_path.name,
                    "page": page_number,
                    "text": text
                })

    if not documents:
        raise FileNotFoundError("No readable PDF documents were found.")

    return documents

def chunk_documents(documents):
```

```

chunks = []

for item in documents:
    text = item["text"]
    start = 0

    while start < len(text):
        end = min(start + CHUNK_SIZE, len(text))
        chunk_text = text[start:end].strip()

        if chunk_text:
            chunks.append({
                "source": item["source"],
                "page": item["page"],
                "text": chunk_text
            })

        if end >= len(text):
            break

        start = end - CHUNK_OVERLAP

return chunks

def build_vector_store(chunks, embedder):
    texts = [item["text"] for item in chunks]

    embeddings = embedder.encode(
        texts,
        normalize_embeddings=True,
        show_progress_bar=True
    )

    embeddings = np.asarray(embeddings, dtype="float32")

    index = faiss.IndexFlatIP(embeddings.shape[1])
    index.add(embeddings)

    return index

def retrieve(query, index, chunks, embedder, top_k=TOP_K):
    query_embedding = embedder.encode(
        [query],
        normalize_embeddings=True
    )

    query_embedding = np.asarray(
        query_embedding,
        dtype="float32"
    )

    scores, positions = index.search(
        query_embedding,
        min(top_k, len(chunks))
    )

    results = []

    for score, position in zip(scores[0], positions[0]):
        if position == -1:
            continue

        results.append({
            "score": float(score),
            "source": chunks[position]["source"],
            "page": chunks[position]["page"],
            "text": chunks[position]["text"]
        })

    return results

```

```

def answer_query(query, index, chunks, embedder):
    out_of_scope_terms = [
        "driving licence",
        "passport",
        "railway ticket",
        "cricket",
        "weather",
        "movie",
        "stock market"
    ]

    if any(term in query.lower() for term in out_of_scope_terms):
        return {
            "answer": "I cannot answer this question because the required information is outside the available government-scheme documents.",
            "sources": []
        }

    retrieved = retrieve(
        query,
        index,
        chunks,
        embedder
    )

    lines = [
        "Answer based only on retrieved government-scheme context:"
    ]

    if not retrieved:
        lines.append(
            "I cannot determine an answer from the available documents."
        )

    for number, item in enumerate(retrieved, start=1):
        lines.append("")
        lines.append("Retrieved Chunk " + str(number))
        lines.append("Source: " + item["source"])
        lines.append("Page: " + str(item["page"]))
        lines.append(
            "Similarity: " + str(round(item["score"], 4))
        )
        lines.append("Retrieved context:")
        lines.append(item["text"])

    return {
        "answer": "\n".join(lines),
        "sources": retrieved
    }

def run_demo(index, chunks, embedder):
    queries = [
        "A farmer family owns cultivable land and wants to know whether PM-KISAN eligibility information applies.",
        "An urban family does not own a pucca house anywhere in India. Which housing scheme information applies?",
        "A landholding farmer family wants information about PM-KISAN eligibility and exclusions.",
        "An urban family with no pucca house wants to know about housing eligibility, while also asking whether a farmer scheme applies.",
        "What is the procedure for getting a driving licence?"
    ]

    for number, query in enumerate(queries, start=1):
        print("\n" + "=" * 80)
        print("QUERY", number)
        print(query)
        print("=" * 80)

        result = answer_query(
            query,
            index,

```

```

        chunks,
        embedder
    )

    print("\nANSWER")
    print(result["answer"])

    print("\nRETRIEVED CHUNKS")

    for rank, item in enumerate(
        result["sources"],
        start=1
    ):
        print("\nChunk", rank)
        print("Source:", item["source"])
        print("Page:", item["page"])
        print("Similarity:", round(item["score"], 4))
        print("Text:", item["text"][:700])

def main():
    print("Government Scheme Eligibility RAG Agent")
    print("Embedding model:", EMBEDDING_MODEL)
    print("Chunk size:", CHUNK_SIZE)
    print("Chunk overlap:", CHUNK_OVERLAP)
    print("Top-k:", TOP_K)

    documents = load_documents()

    print("Pages loaded:", len(documents))

    chunks = chunk_documents(documents)

    print("Chunks created:", len(chunks))

    embedder = SentenceTransformer(
        EMBEDDING_MODEL
    )

    index = build_vector_store(
        chunks,
        embedder
    )

    print("FAISS index created successfully.")

    run_demo(
        index,
        chunks,
        embedder
    )

    print("\n" + "=" * 80)
    print("INTERACTIVE MODE")
    print("Type a citizen profile or question.")
    print("Type exit to stop.")
    print("=" * 80)

    while True:
        query = input(
            "\nEnter citizen profile/question: "
        ).strip()

        if query.lower() == "exit":
            break

        if not query:
            print(
                "Input error: please enter a citizen profile or question."
            )
            continue

```

```

    result = answer_query(
        query,
        index,
        chunks,
        embedder
    )

    print("\n" + result["answer"])

if __name__ == "__main__":
    main()

```

10. Test Queries

Query	Question	Type
1	A farmer family owns cultivable land and wants to know whether PM-KISAN eligibility information applies.	Answerable
2	An urban family does not own a pucca house anywhere in India. Which housing scheme information applies?	Answerable
3	A landholding farmer family wants information about PM-KISAN eligibility and exclusions.	Answerable
4	An urban family with no pucca house wants to know about housing eligibility, while also asking whether a farmer scheme applies.	Multiple information
5	What is the procedure for getting a driving licence?	Out of scope

11. Output

```
=====
QUERY 1
A farmer family owns cultivable land and wants to know whether PM-KISAN eligibility information applies.
=====

ANSWER
Applicable scheme information based only on retrieved document context:

Scheme: PM-KISAN
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 5
Retrieved eligibility/context clause:
Chairman/Chief, authorizing any tribal family to cultivate a piece of land, may be accepted. Such certification of village Chairman/Chief shall be authenticated by the concerned sub-divisional officers. All the exclusions under the Operational Guidelines will be applicable.” 5.5.2 For identification of bona fide beneficiary under PM-Kisan Scheme in Nagaland, the following procedure was approved by the Committee: 1) In case of community owned cultivable land in the state of Nagaland which is under permanent cultivation, for identification of beneficiaries under PM-KISAN, the certificate issued by the village council/authority/village chieftan regarding land holding, duly verified by the administrative head of the circle/sub-division and countersigned by the Deputy Commissioner of the District shall suffice subject to the exclusions under the operational guidelines. 2) In case of cultivabl

RETRIEVED CHUNKS

Chunk 1
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 5
Similarity: 0.724
Text: Chairman/Chief, authorizing any tribal family to cultivate a piece of land, may be accepted. Such certification of village Chairman/Chief shall be authenticated by the concerned sub-divisional officers. All the exclusions under the Operational Guidelines will be applicable.” 5.5.2 For identification of bona fide beneficiary under PM-Kisan Scheme in Nagaland, the following procedure was approved by the Committee: 1) In case of community owned cultivable land in the state of Nagaland which is under permanent cultivation, for identification of beneficiaries under PM-KISAN, the certificate issued by the village council/authority/village chieftan regarding land holding, duly verified by the admin

Chunk 2
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 5
Similarity: 0.7179
Text: 5 iv) In case of transfer of cultivable land for reasons other than death after 01.02.2019, the officials have to reassess the eligibility of the transferor and decide whether the benefits have to be stopped or not and intimate the authorities so that the PM-KISAN portal can be updated. Transferee

=====
QUERY 2
An urban family does not own a pucca house anywhere in India. Which housing scheme information applies?
=====

ANSWER
Applicable scheme information based only on retrieved document context:

Scheme: PMAY-U 2.0
Source: Operational-Guidelines-of-PMAY-U-2.pdf
Page: 16
Retrieved eligibility/context clause:
son, or transgender, the house will be made in the individual’s name. In case of death of beneficiary(s), the legal heir of the beneficiary will get the benefit under the scheme. 1.9 To enable the applicant to avail benefits of PMAY-U 2.0, the inclusion of name of female member of the family in registered title deed/sale deed of the house acquired/purchased during the Mission period, should also be allowed at a later stage and State/UT should make provision for exemption of additional stamp duty and/or registration charges for such cases. 1.10 An undertaking from the beneficiary will be taken as a declaration of the fulfilment of eligibility criteria of income and not owning a pucca house anywhere in India on his/her name or any other family members in a specified format as per Annexure-2 (A, B & C) of the Scheme Guidelines. 1.11 State/UT Governments and Implementing Agencies should ensure

RETRIEVED CHUNKS

Chunk 1
Source: Operational-Guidelines-of-PMAY-U-2.pdf
Page: 16
Similarity: 0.7078
Text: son, or transgender, the house will be made in the individual’s name. In case of death of beneficiary(s), the legal heir of the beneficiary will get the benefit under the scheme. 1.9 To enable the applicant to avail benefits of PMAY-U 2.0, the inclusion of name of female member of the family in registered title deed/sale deed of the house acquired/purchased during the Mission period, should also be allowed at a later stage and State/UT should make provision for exemption of additional stamp duty and/or registration charges for such cases. 1.10 An undertaking from the beneficiary will be taken as a declaration of the fulfilment of eligibility criteria of income and not owning a pucca house anywhere in India on his/her name or any other family members in a specified format as per Annexure-2 (A, B & C) of the Scheme Guidelines. 1.11 State/UT Governments and Implementing Agencies should ensure

Chunk 2
Source: Operational-Guidelines-of-PMAY-U-2.pdf
Page: 15
Similarity: 0.6688
Text: g in Partnership (AHP) iii. Affordable Rental Housing (ARH) iv. Interest Subsidy Scheme (ISS) 1.3 A beneficiary family will comprise of husband, wife, unmarried sons and/or unmarried daughters. Families belonging to EWS/LIG/MIG segments, living in urban areas, should not own a pucca house [(an all
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=====
QUERY 3
A landholding farmer family wants information about PM-KISAN eligibility and exclusions.
=====

ANSWER
Applicable scheme information based only on retrieved document context:

Scheme: PM-KISAN
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 2
Retrieved eligibility/context clause:
2 PRADHAN MANTRI KISAN SAMMAN NIDHI (PM-KISAN) SCHEME OPERATIONAL GUIDELINES 1. Scheme With a view to provide income support to all landholding farmers' families in the country, having cultivable land, the Central Government has implemented a Central Sector Scheme, namely, "Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)". 2. Objective and Benefits The scheme aims to supplement the financial needs of all landholding farmers' families in procuring various inputs to ensure proper crop health and appropriate yields, commensurate with the anticipated farm income as well as for domestic needs. Under the Scheme an amount of Rs.6000/- per year is released by the Central Government online directly into the bank accounts of the eligible farmers under Direct Benefit Transfer mode, subject to certain exclusions. 3. Definition of farmer's family A landholder farmer's family is defined as "a family comprising of a landholder and his/her family members residing in the same household."

RETRIEVED CHUNKS

Chunk 1
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 2
Similarity: 0.6988
Text: 2 PRADHAN MANTRI KISAN SAMMAN NIDHI (PM-KISAN) SCHEME OPERATIONAL GUIDELINES 1. Scheme With a view to provide income support to all landholding farmers' families in the country, having cultivable land, the Central Government has implemented a Central Sector Scheme, namely, "Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)". 2. Objective and Benefits The scheme aims to supplement the financial needs of all landholding farmers' families in procuring various inputs to ensure proper crop health and appropriate yields, commensurate with the anticipated farm income as well as for domestic needs. Under the Scheme an amount of Rs.6000/- per year is released by the Central Government online directly into the bank accounts of the eligible farmers under Direct Benefit Transfer mode, subject to certain exclusions. 3. Definition of farmer's family A landholder farmer's family is defined as "a family comprising of a landholder and his/her family members residing in the same household."

Chunk 2
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 5
Similarity: 0.6593
Text: 5 iv) In case of transfer of cultivable land for reasons other than death after 01.02.2019, the officials have to reassess the eligibility of the transferee and decide whether the benefits have to be stopped or not and intimate the authorities so that the PM-KISAN portal can be updated. Transferee has to submit a declaration in a specified format as per Annexure-2 (A, B & C) of the Scheme Guidelines. 1.11 State/UT Governments and Implementing Agencies should ensure that the information entered in the portal is correct and up-to-date.

Enter citizen profile/question: A landholding farmer family wants information about PM-KISAN eligibility and exclusions.
Applicable scheme information based only on retrieved document context:

Scheme: PM-KISAN
Source: RevisedPM-KISANOperationalGuidelines(English).pdf
Page: 2
Retrieved eligibility/context clause:
2 PRADHAN MANTRI KISAN SAMMAN NIDHI (PM-KISAN) SCHEME OPERATIONAL GUIDELINES 1. Scheme With a view to provide income support to all landholding farmers' families in the country, having cultivable land, the Central Government has implemented a Central Sector Scheme, namely, "Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)". 2. Objective and Benefits The scheme aims to supplement the financial needs of all landholding farmers' families in procuring various inputs to ensure proper crop health and appropriate yields, commensurate with the anticipated farm income as well as for domestic needs. Under the Scheme an amount of Rs.6000/- per year is released by the Central Government online directly into the bank accounts of the eligible farmers under Direct Benefit Transfer mode, subject to certain exclusions. 3. Definition of farmer's family A landholder farmer's family is defined as "a family comprising of a landholder and his/her family members residing in the same household."

Enter citizen profile/question: An urban family with no pucca house wants to know about housing eligibility, while also asking whether a farmer scheme applies.
Applicable scheme information based only on retrieved document context:

Scheme: PMAY-U 2.0
Source: Operational-Guidelines-of-PMAY-U-2.pdf
Page: 18
Retrieved eligibility/context clause:
assessment of housing demand under different verticals through suitable means for assessing the actual demand of housing as per Annexure-3. The beneficiaries can register themselves for housing demand with all details on unified Web-portal. The beneficiaries will be validated by States/UTs/ULBs as per the eligibility criteria of the scheme guidelines. States/UTs/ULBs should verify and exclude the beneficiary families from the list who are having a pucca house in rural area and migrated to city to take advantage of PMAY-U 2.0. 4.5 States/UTs/ULBs should also identify the beneficiaries from the Special Focus Group (SFG), as mentioned in para 3.4, for inclusion in PMAY-U 2.0 and accordingly provide benefits to these groups on priority. 4.6 Based on demand assessment and other accessible data, cities will complete the online format of the City-wise Housing Plan on the Unified Web-portal in the format of the City-wise Housing Plan.

Enter citizen profile/question: What is the procedure for getting a driving licence?
I cannot answer this question because the required information is outside the available government-scheme documents.

```

12. Observation

The system loads the available government PDFs, extracts page-level text, divides the text into overlapping chunks, generates sentence embeddings, indexes the embeddings using FAISS, and retrieves the most similar chunks for each query. The retrieved source and page information provide evidence for the response.

13. Grounding and Out-of-Scope Handling

The agent displays the retrieved document chunks used for the response. For an out-of-scope question such as a driving-licence procedure, the program is designed to refuse because the required information is outside the available government-scheme documents.

14. Result

A RAG-based Government Scheme Eligibility Agent was implemented using PDF document loading, overlapping text chunking, sentence embeddings and FAISS similarity retrieval. The program demonstrates retrieval for government-scheme eligibility questions and displays supporting source and page information.

15. Conclusion

The experiment demonstrates how a RAG pipeline can retrieve relevant information from government scheme documents and provide grounded responses with supporting context. This reduces reliance on unsupported answers and makes the retrieved evidence visible to the user.

16. Execution Details

Pages loaded in the demonstrated run: 119

Chunks created in the demonstrated run: 338

Embedding model: sentence-transformers/all-MiniLM-L6-v2

Vector store: FAISS

Chunk size: 900 characters

Chunk overlap: 150 characters

Top-k: 5